MAS1004 Data & AI, 2026 Fall

Assignment 2: Long report

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Leaderboard nickname: yosam. My last release of each track is `track-a-1` (Track A: the starter recipe on all training photos, from commit `ff34d66`) and `track-b-1` (Track B: the model called b1). The leaderboard scores next to my validation scores are a note for tomorrow: [the leaderboard score of `track-a-1` and of `track-b-1` next to the validation scores 0.4942 / 0.4971 (Track A) and 0.4835 (Track B), read from the leaderboard in the morning — not written yet].

What I saw in the photos

These are my notes from 9 November, written after I looked at the training photos.

Busan. Because it is a port city, I see things like seafood markets, the coast, the port, or Korean signs. But a gymnasium with our national flag on it, or a school playground where an apartment name is slightly visible, cannot be told apart from Seoul.

Osaka. In Osaka too, you could tell it is Japanese by looking at things like inns. But the coast or the sea — if you know the Japanese feel you can tell, but in fact I think it is hard to tell apart from Busan. And there are really a lot of strange natural things. I do not really know why either.

Seoul. Seoul, as befits the capital of the Joseon Dynasty long ago, does seem to have many cultural sites. There are many scenes of Gyeongbokgung, and many people wearing traditional clothing.

Tokyo. Tokyo too can be distinguished if you know the Japanese style, but there are also many parts where I cannot really tell which area of Osaka it is.

Taipei. Taipei looks different from our East Asia, but I do not know what to call it.

Busan and Osaka are the two cities with the lowest per-city scores in every run: in the baseline (seed 0) they are 0.3914 and 0.3228, and in Track B's b1 they are 0.3791 and 0.3369. That goes with what I wrote above, where I found Busan and Osaka hard to tell apart.

Track A

tag	what changed from the run before	validation
`baseline`, seed 0	nothing: the starter recipe on all 44,257 training photos	0.4942
`baseline`, seed 1	the same recipe and the same photos; only the seed changed	0.4971
`no_flip`, seed 0	the left-right flip removed from `train_transform`	0.4926
`drop_nonstreet`, seed 0	the flip back, so the recipe is the starter recipe again, on a selection of 44,172 photos instead of 44,257: 85 dropped as not a street scene	0.4891

Every number in that table is from `experiments.csv`, which is printed in full under "All runs". All four runs went through `src/official.py` with `rules.TRACK_A_BUDGET`, 249,984 photos each, on Colab's GPU.

Why I made each change, and what I think the result means (my own words, from `LOG.md`):

baseline, seeds 0 and 1. Why: to have a reference, and to see how much the score moves by chance when only the seed changes. What it means: the two seeds differ by about 0.003, so a change has to move the score by more than that before I believe it.

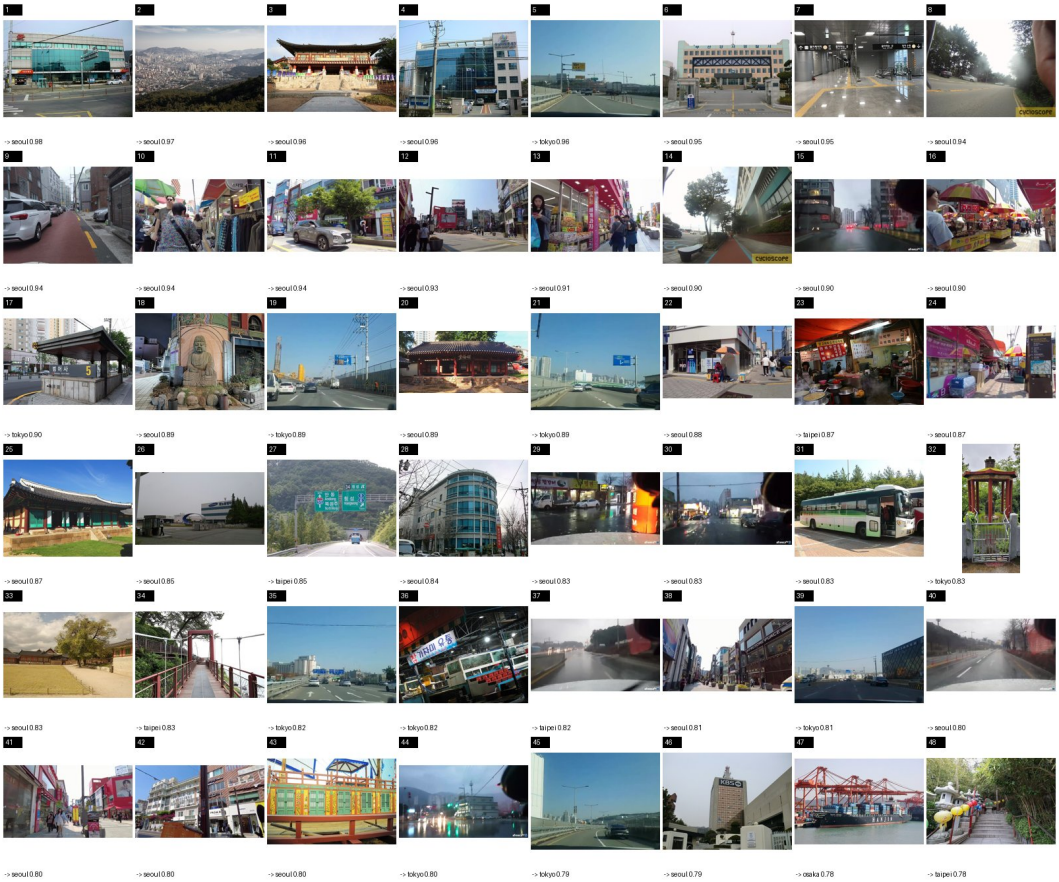
no_flip. Why: a flip turns signs into mirror writing and puts traffic on the wrong side, and Japan drives on the left while Korea and Taiwan drive on the right, so I thought it destroys clues. What it means: 0.4926 is within the seed noise of baseline, so removing the flip did not help; maybe signs and traffic direction are a small part of most photos.

drop_nonstreet. Why: my rule, written before I chose which photos to look at — drop training photos you could not see while walking a street: nature only, inside a room (museum, restaurant, office), exhibits, food, maps, a plane in the sky; keep walkable covered spaces such as shopping arcades, underground malls and stations; street photos that are just hard to tell from another city stay in. What it means: as I predicted, removing 85 photos (0.2%) did not help. It came out about 0.005 below baseline, but with only one seed and per-city swings of 0.02–0.04 between seeds, I cannot tell whether that is a real drop or noise. To see a real effect I would have to apply the rule to all training photos, not only the 48 most doubtful per city.

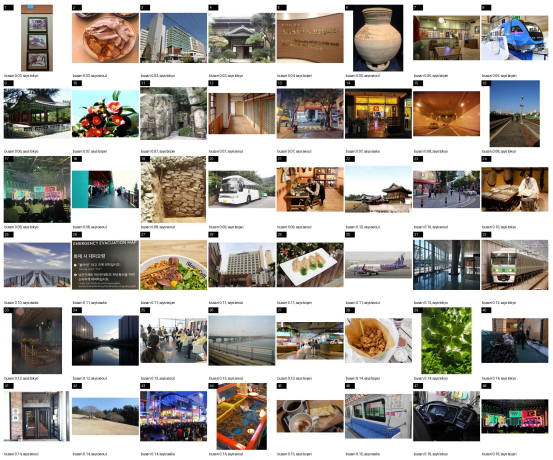
My final recipe and selection. The recipe is the starter recipe: in `experiments.csv` the recipe hash of `drop_nonstreet` (`17ac485223fa`) is the same as both `baseline` runs, because the left-right flip is back and nothing else in `track_a/recipe.py` differs. The selection is `track_a/selection.csv`, 44,172 of the 44,257 training photos, which validated at 0.4891. The run that is actually released is `track-a-1`, the starter recipe on all training photos, with validation 0.4942 (seed 0) and 0.4971 (seed 1). Its leaderboard score next to those two numbers: [to be filled in from tomorrow's leaderboard — not written yet].

Decisions you made from photos

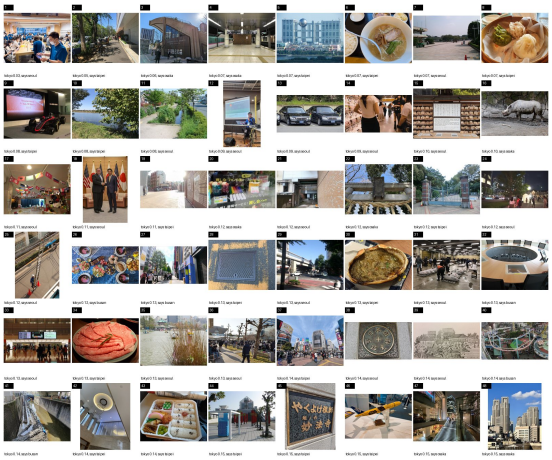
Two kinds of sheets were drawn from the `baseline` seed 0 model on Colab, both numbered so that a photo can be named by its sheet and its number: `curate.py mistakes` (`--every 2`, 6,039 validation photos scored, showing the ones the model got wrong most confidently) and `curate.py suspects` (`--every 4`, 11,065 training photos scored, showing the 48 photos per city with the lowest probability for their own city). All the counts in this section are from those two runs' output in `results/colab_logs/`.



Busan, `mistakes` sheet: the 48 validation photos baseline seed 0 got wrong most confidently. Of the 695 Busan validation photos read with `--every 2`, 430 were wrong.



Busan, `suspects` sheet: the 48 Busan training photos with the lowest Busan probability. I dropped 24 of the 48 on this sheet.



Tokyo, `suspects` sheet: the same for Tokyo. I dropped 22 of the 48 on this sheet.

The rule I wrote before choosing which photos to drop. Drop training photos you could not see while walking a street — nature only, inside a room (museum, restaurant, office), exhibits, food, maps, a plane in the sky. Keep walkable covered spaces such as shopping arcades, underground malls and stations. Street photos that are just hard to tell from another city stay in.

I applied that rule to the five `suspects` sheets of `baseline` seed 0: osaka, tokyo, seoul, busan and taipei.

What it changed. 85 photos left the selection (osaka 14, tokyo 22, seoul 12, busan 24, taipei 13, which is the difference between the two columns of the count below), so `track_a/selection.csv` went from 44,257 rows to 44,172 rows. Every dropped photo and the reason are in `track_a/dropped.csv`. The photos are not deleted, only left out, so `curate.py undrop` can put any of them back. The scores before and after: 0.4942 (baseline, seed 0) and 0.4971 (baseline, seed 1) before, 0.4891 after, with the same recipe. The output of `curate.py count` on the new selection:

```
city      selected     of   by source
seoul        11262  11274   commons 6758, osv5m 320, youtube 4184
busan         5167   5191   commons 3464, youtube 1703
tokyo        13325  13347   commons 6954, osv5m 1916, youtube 4455
osaka         7186   7200   commons 5335, osv5m 341, youtube 1510
taipei        7232   7245   commons 5656, osv5m 533, youtube 1043
total 44172 rows
dropped: 85 not a street scene
```

The requests I gave my agent for these decisions (step 7 of my work with it, copied exactly as I sent it):

```
My rule, written before choosing: drop training photos you could not see while walking a street -
↪  nature only, inside a room (museum, restaurant, office), exhibits, food, maps, a plane in the
↪  sky. Keep walkable covered spaces such as shopping arcades, underground malls and stations.
↪  Street photos that are just hard to tell from another city stay in. I applied it to all five
↪  suspect sheets of baseline seed 0.

Drop photos 1, 6, 10, 11, 20, 33, 34, 36, 37, 39, 41, 46, 47, 48 from
↪  suspects_baseline_seed0_osaka_00, reason "not a street scene".
Drop photos 1, 6, 8, 9, 12, 13, 14, 16, 17, 18, 26, 28, 30, 31, 32, 34, 38, 39, 42, 43, 45, 46 from
↪  suspects_baseline_seed0_tokyo_00, reason "not a street scene".
Drop photos 4, 12, 14, 20, 21, 30, 36, 37, 39, 41, 43, 45 from suspects_baseline_seed0_seoul_00,
↪  reason "not a street scene".
Drop photos 1, 2, 6, 7, 8, 10, 12, 17, 21, 24, 26, 27, 29, 30, 33, 35, 37, 38, 39, 44, 45, 46, 47,
↪  48 from suspects_baseline_seed0_busan_00, reason "not a street scene".
```

```
Drop photos 1, 2, 4, 10, 11, 14, 15, 19, 21, 32, 38, 47, 48 from suspects_baseline_seed0_taipei_00,
↪  reason "not a street scene".

Then put the left-right flip back in track_a/recipe.py so the recipe is the same as baseline,
↪  change nothing else, and show me the diff. Train the new selection on Colab with the tag
↪  drop_nonstreet, and tell me when to press Run all.
```

Track B

b1, the first step. `train_b.py` as it came: `resnet18` with its ImageNet weights, 224 px, 4 epochs, batch 64, no balancing, on every photo in `data/starter/train`, submitted with `--submit`. It validated at 0.4835 (seoul 0.591, busan 0.3791, tokyo 0.5986, osaka 0.3369, taipei 0.5118) after 176,896 photos through the network, from the b1 row of `experiments.csv`; the run's own output is in `results/colab_logs/`.

Why: to have a valid Track B submission and a reference for later Track B changes. What it means: b1 uses the same model, image size and augmentation as the Track A baseline, but 4 epochs is only about 177,000 photos instead of 250,000. Its score is about 0.01 below baseline, more than the 0.003 seed noise, so I think it is lower mainly because it trained on fewer photos.

b2 did not run. The next step was the same file with 6 epochs instead of 4 — `python src/train_b.py --tag b2 --epochs 6 --submit` followed by `python src/check_submission.py --no-val`, queued in `colab_runs.txt`. The run never started: the free Colab GPU limit was reached, so Colab would not give me a GPU that day. There is therefore no b2 row in `experiments.csv`, no b2 validation score, and `submission/` still holds b1. Nothing else about b2 was changed: `train_b.py` is identical to the file b1 used, and 6 epochs instead of 4 is the only difference.

Extra photos. None. Both Track B runs used only `data/starter/train`, so there was nothing to collect from a phone or put in Google Drive.

My validation score next to my leaderboard score. b1 validated at 0.4835, and `track-b-1` is the released model. [its leaderboard score next to 0.4835, and why the two might disagree, in my own words — to be read from the leaderboard tomorrow and written then].

Your submission, as the grader sees it

```
C:\Users\gyuhw\Documents\mas1004-assignment2\submission\model.onnx: 44.7 MB, 11.17 million
↪  parameters, 1.81 billion multiply-adds per 224 x 224 photo
  limits: 12 million parameters, 2.0 billion multiply-adds
  format and limits: OK
```

The description of the model that `train_b.py` wrote next to it, in `submission/submission.json`:

```
{
  "image_size": 224,
  "starting_weights": "torchvision resnet18 IMAGENET1K_V1",
  "data": "data/starter/train",
  "summary": "resnet18 at 224, 4 epochs"
}
```

What made the difference

Neither Track A change I measured beat the baseline. With the starter recipe on all training photos the score was 0.4942 (seed 0) and 0.4971 (seed 1); removing the left-right flip gave 0.4926, and taking the 85 non-street photos out of the selection gave 0.4891. Track B never got past its reference run: b1 validated at 0.4835 on 176,896 photos, and b2, the same file with 6 epochs, could not run because the

free Colab GPU limit was reached. Every number here is a `validation` value from `experiments.csv`.

[Which change helped most and which least, in my own words, and what I would do next with one more week — not written yet.]

My plan for Track A, from `LOG.md`, is to apply the photo rule to all training photos, not only the 48 most doubtful per city, because 85 photos out of 44,257 was too small a change to see through the seed noise. The 6-epoch Track B run for b2 is still queued in `colab_runs.txt` and has not run yet.

All runs

```
finished,track,tag,seed,official_budget,budget,val_score,seoul,busan,tokyo,osaka,taipei,train_seco
↪  nds,device,selection,recipe_sha,selection_sha,commit
2026-11-09T11:45:26,A,baseline,0,True,249984,0.4942,0.5717,0.3914,0.6406,0.3228,0.5444,2262.5,cuda
↪  ,all training
↪  photos,17ac485223fa,,1c3f89d
2026-11-09T12:23:53,A,baseline,1,True,249984,0.4971,0.5795,0.3921,0.6176,0.368,0.5283,2198.3,cuda,
↪  all training
↪  photos,17ac485223fa,,1c3f89d
2026-11-09T12:52:51,B,b1,,,176896,0.4835,0.591,0.3791,0.5986,0.3369,0.5118,1634.3,cuda,resnet18
↪  224px; data/starter/train,,,1c3f89d
2026-11-09T15:22:41,A,no_flip,0,True,249984,0.4926,0.5828,0.3827,0.6145,0.3492,0.5337,2176.8,cuda,
↪  all training
↪  photos,d555940f08fe,,1c70c1a
2026-11-09T22:53:38,A,drop_nonstreet,0,True,249984,0.4891,0.5815,0.3799,0.6236,0.3477,0.5127,2227.
↪  2,cuda,"selection.csv, 44172
↪  rows",17ac485223fa,555bf2406914,f3e5e3f
```

Lab notebook

```
# Log

One entry for each recipe, selection or Track B model you train, newest at
the bottom. The numbers come from `experiments.csv`. "Why" and "what it means"
are in your own words: your agent asks you for them and does not make them up.
End the log with your plan for what to try next.

<!-- The form of an entry:

## <date> <tag> (Track A or B)
- changed: what is different from the run before
- why: your reason
- validation: 0.000 (before: 0.000)
- what it means: what you think the result says
-->

## 2026-11-09 baseline (Track A)

- changed: nothing. The starter recipe on all training photos (44,257), first with seed 0 and then
↪  with seed 1.
- why: to have a reference, and to see how much the score moves by chance when only the seed
↪  changes.
- validation: 0.4942 with seed 0 and 0.4971 with seed 1 (seed 0 per city: seoul 0.5717, busan
↪  0.3914, tokyo 0.6406, osaka 0.3228, taipei 0.5444).
- what it means: the two seeds differ by about 0.003, so a change has to move the score by more
↪  than that before I believe it.

## 2026-11-09 b1 (Track B)

- changed: `train_b.py` as it came - resnet18 at 224 px, 4 epochs, batch 64, no balancing, on
↪  every photo in `data/starter/train`.
- why: to have a valid Track B submission and a reference for later Track B changes.
```

```
- validation: 0.4835 (seoul 0.591, busan 0.3791, tokyo 0.5986, osaka 0.3369, taipei 0.5118).
- what it means: b1 uses the same model, image size and augmentation as the Track A baseline, but
↪  4 epochs is only about 177,000 photos instead of 250,000. Its score is about 0.01 below
↪  baseline, more than the 0.003 seed noise, so I think it is lower mainly because it trained on
↪  fewer photos.

## 2026-11-09 no_flip (Track A)

- changed: removed the left-right flip (`T.RandomHorizontalFlip()`) from `train_transform`. The
↪  rest of the starter recipe is unchanged and the photos are still all training photos, so this
↪  is baseline without the flip.
- why: a flip turns signs into mirror writing and puts traffic on the wrong side, and Japan drives
↪  on the left while Korea and Taiwan drive on the right, so I thought it destroys clues.
- validation: 0.4926 (before: 0.4942 and 0.4971 from baseline seeds 0 and 1); seoul 0.5828, busan
↪  0.3827, tokyo 0.6145, osaka 0.3492, taipei 0.5337.
- what it means: 0.4926 is within the seed noise of baseline, so removing the flip did not help;
↪  maybe signs and traffic direction are a small part of most photos.

## 2026-11-09 drop_nonstreet (Track A)

- changed: the starter recipe again (the left-right flip is back, nothing else in `recipe.py`
↪  differs from `baseline`) on a smaller selection. 85 training photos dropped with the reason
↪  "not a street scene": osaka 14, tokyo 22, seoul 12, busan 24, taipei 13, taken from the five
↪  suspect sheets of baseline seed 0. `selection.csv` went from 44,257 to 44,172 rows.
- why: my rule, written before I chose which photos to look at: drop training photos you could not
↪  see while walking a street - nature only, inside a room (museum, restaurant, office),
↪  exhibits, food, maps, a plane in the sky. Keep walkable covered spaces such as shopping
↪  arcades, underground malls and stations. Street photos that are just hard to tell from another
↪  city stay in.
- expected before seeing the result: little or no change, because 85 photos are only about 0.2% of
↪  the training set, and I only looked at the 48 most doubtful photos per city from every fourth
↪  training photo.
- validation: 0.4891 (before: 0.4942 and 0.4971 from baseline seeds 0 and 1); seoul 0.5815, busan
↪  0.3799, tokyo 0.6236, osaka 0.3477, taipei 0.5127.
- what it means: as I predicted, removing 85 photos (0.2%) did not help. It came out about 0.005
↪  below baseline, but with only one seed and per-city swings of 0.02-0.04 between seeds, I
↪  cannot tell whether that is a real drop or noise. To see a real effect I would have to apply
↪  the rule to all training photos, not only the 48 most doubtful per city.

## Plan

- drop_nonstreet: apply the rule to all training photos, not only the 48 most doubtful per city.
```